Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

Odyssey2024 epsfig,amssymb,amsmath

Paper Template for Odyssey 2024

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This is a layout specification and template definition for the papers of Odyssey 2024 Conference, which will be held in Quebec, Canada, Jun 18 – 21, 2024. This template has evolved from the Odyssey 2014 template that traces back to Eurospeech 2001 and aims at producing conference proceedings in electronic form. The format is essentially the one used for the IEEE ICASSP conferences, but using the standard A4 paper size.

1 Introduction

This template can be found on the conference website. Please use either one of the template files found on this website when preparing your submission. If there are special questions or wishes regarding paper preparation and submission for Odyssey 2024, correspondence should be addressed to <info@odyssey2024.org>.

Information for full paper submission is available on the web at <http://www.odyssey2024.org> under which you also will find instructions for paper preparation and usage of templates.

2 Page layout and style

The page layout should match with the following rules. A highly recommended way to meet these requirements is to use one of the templates provided and to check details against this example file. If for some reason you cannot use any of the templates, please follow these rules as carefully as possible, or contact the organizers at <info@odyssey2024.org> for further instructions.

2.1 Basic layout features

- Proceedings will be printed in A4 format. The layout is designed so that the papers, when printed in US Letter format, will include all material but the margins will not be symmetric. PLEASE TRY TO MAKE YOUR SUBMISSION IN A4 FORMAT, if possible, although this is not an absolute requirement.
- Two columns are used except for the title part and possibly for large figures that need a full page width.
- Left margin is 20 mm.
- Column width is 80 mm.
- Spacing between columns is 10 mm.
- Top margin 25 mm (except first page 30 mm to title top).
- Text height (without headers and footers) is maximum 235 mm.
- Headers and footers must be left empty (they will be added for printing).
- Check indentation and spacing by comparing to this example file (in PDF).

2.1.1 Headings

Section headings are centered in boldface with the first word capitalized and the rest of the heading in lower case. Sub-headings appear like major headings, except they start at the left margin in the column. Sub-sub-headings appear like sub-headings, except they are in italics and not boldface. See examples in this file. No more than 3 levels of headings should be used.

2.2 Fonts

The font used for the main text is Times. The recommended font size is 9 points which is also the minimum allowed size. Other font types may be used if needed for special purposes. Remember, however, to embed all the fonts in your final PDF file!

LaTeX users: DO NOT USE THE Computer Modern FONT FOR TEXT (Times is specified in the style file). If possible, make the final document using POSTSCRIPT FONTS since, for example, equations with non-PS Computer Modern are very hard to read on screen.

2.3 Figures

All figures must be centered in the column (or page, if the figure spans both columns). Figure captions should follow each figure and have the format given in Fig. 1.

Figures should preferably be line drawings. If they contain gray levels or colors, they should be checked to print well on a high-quality non-color laser printer. If some figures contain bitmap images, please ensure that their resolution is high enough to preserve readability.

2.4 Tables

An example of a table is shown as Table 1. Somewhat different styles are allowed according to the type and purpose of the table. The caption text may be above or below the table.

Tabel 1: *This is an example of a table.*

ratio	decibels
1/1	0
2/1	≈ 6
3.16	10
10/1	20
1/10	-20
100/1	40
1000/1	60

2.5 Equations

Equations should be placed on separate lines and numbered. Examples of equations are given below. Particularly,

$$x(t) = s(f_\omega(t)) \tag{1}$$

where $f_\omega(t)$ is a special warping function

$$f_\omega(t) = \frac{1}{2\pi j} \oint_C \frac{\nu^{-1k} d\nu}{(1 - \beta\nu^{-1})(\nu^{-1} - \beta)} \tag{2}$$

A residue theorem states that

$$\oint_C F(z)dz = 2\pi j \sum_k Res[F(z), p_k] \tag{3}$$

Applying (3) to (1), it is straightforward to see that

$$1 + 1 = \pi \tag{4}$$

Finally we have proven the secret theorem of all speech sciences. No more math is needed to show how useful the result is!

2.6 Page numbering

Final page numbers will be added later to the document electronically. *Please do not include any headers or footers!*

2.7 References

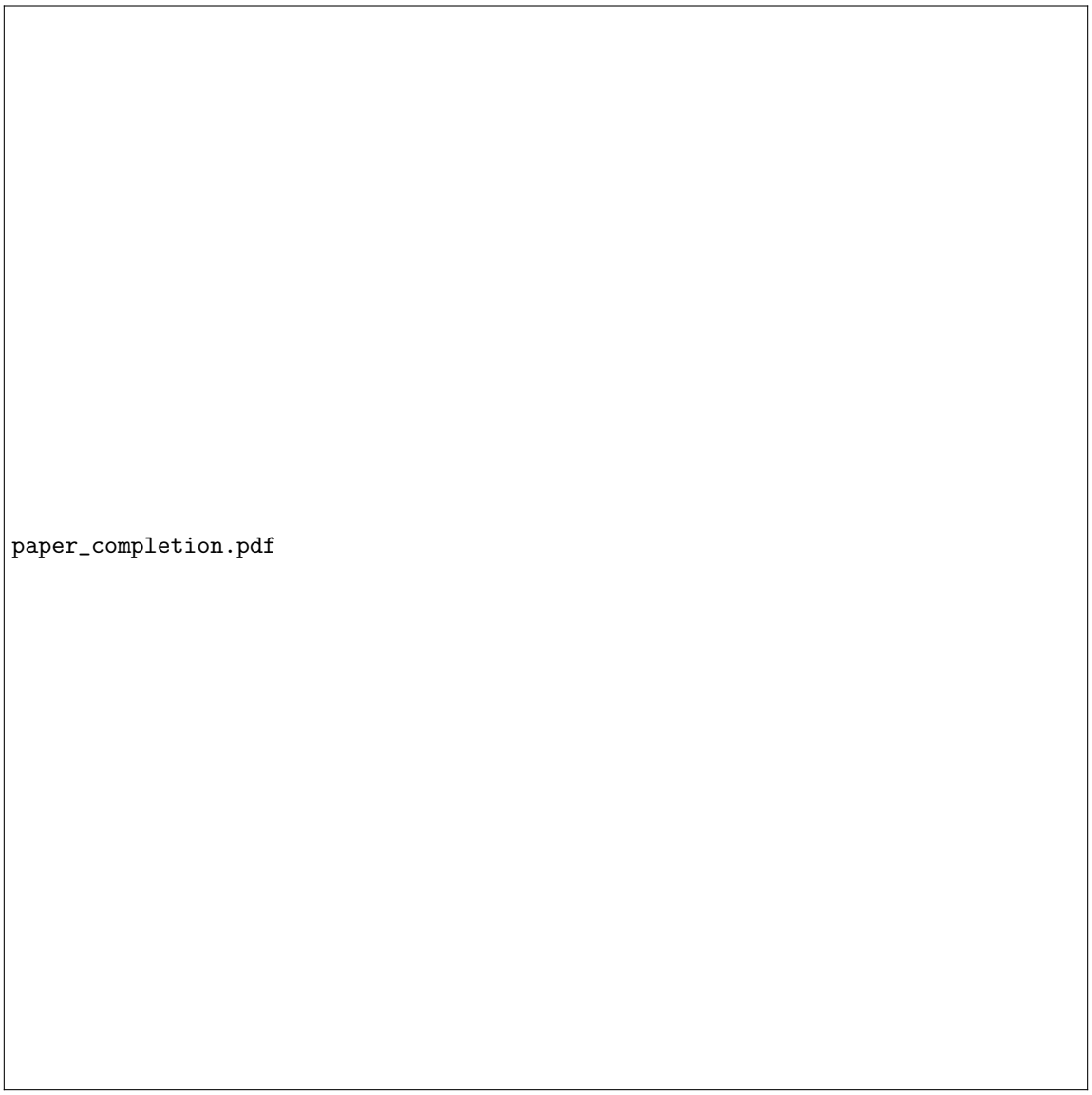
The reference format is the standard IEEE one. References should be numbered in order of appearance, for example [**?**], and later this one [**?**], and even later these other three [**?**, **?**, **?**].

3 Discussion

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4 Conclusion

This template can be found on the conference website <http://www.odyssey2024.org>.



Gambar 1: *This is an example of a figure.*